

1. $n^{\frac{5}{3}} = k^{\frac{2}{5}}$ and $n^{4a+1} = k$, what is the value of a ?

A) $19/24$ B) $5/24$ C) $21/24$ D) $17/24$

2. $3x^2 + bx - 112 = (ax + m)(x - l)$. Given that a , m , and l are integers, which of the following must be true?

A) m is a factor of b
 B) a is a factor of 112
 C) a is a factor of b
 D) l is a factor of 112

3. The perimeter of a right isosceles triangle is $76 + 76\sqrt{2}$. What is the length of the hypotenuse?

A) 76 B) $76\sqrt{2}$ C) 152 D) 114

4.

Sample	Percent in favor	Margin of error
A	67.83%	7.8%
B	72.94%	4.5%

The results of two random samples of votes for a proposition are given in the table. The samples were selected from the same population, and the margins of error were calculated using the same method. Which of the following is the most appropriate reason that the margin of error for sample A is greater than the margin of error for sample B?

- A) Sample A had a larger sample size.
 B) Sample A had a smaller sample size.
 C) Sample A had a smaller number of votes that could not be recorded.
 D) Sample A had a lower percent of favorable responses.

5. 360 is $p\%$ of 10. What is the value of p ?

A) 3600 B) 2400 C) 360 D) 240

6. The vertex of the function $f(x)$ is given as $(9, -13)$, and the point $(3, -4)$ lies on the parabola. If $g(x) = 4f(x - 2)$, what is the value of $g(0) - f(0)$?

A) 61.75 B) 71.25 C) 82.15 D) 91.25

7.

$$-377x + k = 377x + k$$

If k is a constant, how many solutions does the given equation have?

- A) No solutions
 B) One solution
 C) Two solutions
 D) Infinite solutions

8.

The density of a certain type of wood is 770 kilograms per cubic meter. A sample of this type of wood is in the shape of a cube, where each edge has a length of 0.7 meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

- A) 264
 B) 377
 C) 1,571
 D) 2,245

9. The graph of the linear equation passes through the points $(-21, -\frac{89}{2})$ and $(12, \frac{65}{2})$. The graph also passes through the point $(\frac{27}{4}, a)$. What is the value of a ?

A) 20.25 B) 21.85 C) 31.15 D) 32.25

10. In $\triangle ABC$, $\angle B$ is a right angle and the length of $\overline{AB}$ is 456 millimeters. If the $\cos C = \frac{7}{25}$, what is the length, in millimeters, of $\overline{BC}$?
A) 133 B) 124 C) 136 D) 142
11. In triangle ABC and triangle DEF , sides BC and EF each have a side length of 37 inches, and angles B and E each have an angle measure of 63° . Which of the following additional pieces of information is(are) sufficient to prove whether triangle ABC is congruent to triangle DEF ?
- The measure of angles A and C are equal.
 - The length of sides AC and DF are equal.
 - The measures of angles A and D are equal.
- A) I is sufficient, but II and III are not.
B) II is sufficient, but I and III are not.
C) III is sufficient, but I and II are not.
D) II and III are sufficient, but I is not.
12. $77x^2 + (7j + 11k)x + jk = 0$
In the given equation, j and k are positive constants. The product of the solutions to the given equation is ajk , where a is a constant. What is the value of a ?
A) $1/77$ B) $1/154$ C) $1/7$ D) $1/11$
13. The number j is 2,400% greater than the number k . The number k is 350% greater than 8. What is the value of j ?
A) 900 B) 1800 C) 450 D) 1350
14. A square is inscribed in a circle with the equation $x^2 + y^2 - 8x + 10y = 27$. What is the area of the square?
A) 136 B) 144 C) 169 D) 196
15. The equation below relates distinct positive real numbers a , b and c . Which equation correctly expresses c in terms of a and b ?
- $$a = 19b \cdot \sqrt[5]{\left(\frac{c}{20}\right)^4}$$
- A) $\frac{20}{19b} (a)^{\frac{5}{4}}$
B) $\frac{20}{19b} (a)^{\frac{4}{5}}$
C) $20 \left(\frac{a}{19b}\right)^{\frac{5}{4}}$
D) $20 \left(\frac{a}{19b}\right)^{\frac{4}{5}}$
16. Triangle ABC is inscribed in a circle with a radius of 85. The length of AC is 170 and the length of BC is 168. What is the value of $\frac{AB}{BC}$?
A) $13/84$ B) $1/4$ C) $17/84$ D) $25/84$
17. For the exponential function f , the value of $f(2)$ is k , where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?
- A) $f(x) = 729(3)^{x+2}$
B) $f(x) = 6,561(3)^x$
C) $f(x) = 177,147(3)^{x-3}$
D) $f(x) = 59,049(3)^{x-2}$

18. Two numbers r and s are each greater than zero, and the fourth root of r is equal to the seventh root of s . For what value of x is r^{4x-5} equal to s ?

A) 27/16 B) 16/27 C) 36/16 D) 42/16

19. The given equation $ax^2 + 98x + c$ has at least 1 real root and a factor of $kx + j$. What is the greatest possible value of ac ?

A) 2401 B) 2400 C) 2402 D) 2403

20.
$$7kx + 13my = 20.5$$
$$6kx + 5my = -48$$

In the given system of equations, k and m are constants. The system has a solution of $(2, y)$. What is the value of k ?

A) -11 B) -13 C) -15 D) -18

- 21.

x	$g(x)$
4	9
2	0

The table gives two values of x and their corresponding values of $g(x)$, where $g(x) = \frac{f(x)+5}{x+8}$ and f is a linear function. What is the y -coordinate of the y -intercept of the graph $y = f(x)$ in the xy -plane?

A) -113 B) -115 C) -108 D) -117

22. What is the value of $\sin(x^\circ)(\cos 90 - x^\circ) + \sin(90 - x^\circ)(\cos x^\circ)$?

A) 0.5
B) 1
C) 2
D) 90

23. There are red, blue, green, and black marbles in a bag. There are 26 black marbles, 8 green marbles, and 26 blue and red marbles. Given that a randomly selected marble is not black, the probability of selecting a red marble is $\frac{5}{17}$. What is the number of blue marbles in the bag?

A) 16 B) 18 C) 20 D) 24

24. A gardener is purchasing topsoil for a new garden. The garden is in the shape of a rectangular prism. The dimensions of the garden are 54 feet by 192 feet by 2 feet. If it costs \$96 per cubic yard of topsoil, how much does the gardener need to spend in dollars? (1 yard = 3 feet)

A) 73728 B) 76884 C) 71298 D) 78864

25. In the xy -plane, which of the following points lies on a circle with the equation $(x + k)^2 + (y - j)^2 = 841$?

A) $(k + 20, j + 21)$
B) $(k - 20, j - 21)$
C) $(-k + 21, j + 20)$
D) $(-k - 21, -j - 20)$

26. Which of the following inequalities gives all possible values of k where the equation below has no solution?

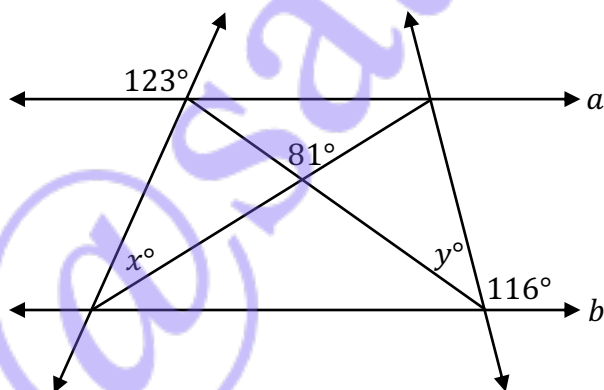
$$4 - 4 \left| \frac{7}{8} + \frac{4}{5}x \right| = \frac{57}{12} + k$$

- A) $k < -\frac{3}{4}$
 B) $k > -\frac{3}{4}$
 C) $k \leq \frac{3}{4}$
 D) $k \geq \frac{3}{4}$

27. A table that weighs 348 pounds puts equal amounts of pressure on each of its 4 identical legs. If the pressure that each leg exerts on the floor is 12 pounds per square inch, what is the area, in square inches, of the bottom of each leg that is touching the floor?

- A) $29\frac{1}{4}$ B) $27\frac{1}{6}$ C) $31\frac{1}{4}$ D) $33\frac{1}{4}$

28. In the figure below, lines a and b are parallel. What is the value of $x + y$?
 A) 22 B) 24 C) 26 D) 28



Note: Figure not drawn to scale

29. The functions f and g are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$?

- I. $f(x) = 232(0.4)^{x+2}$
 II. $g(x) = 232(0.4)(0.4)(0.4)^{x-2}$

- A) I only
 B) II only
 C) I and II
 D) Neither I nor II

30. An equilateral triangle with side lengths of k is inscribed in a circle. Which of the following defines the radius, r , in terms of k ?

- A) $\frac{k}{2}$
 B) $\frac{k\sqrt{3}}{6}$
 C) $\frac{k\sqrt{3}}{3}$
 D) $\frac{k\sqrt{3}}{2}$

31. The function f is defined by $f(x) = 3^x$. The function g is an increasing linear function. In the xy -plane, the graphs of $y = f(x)$ and $y = g(x)$ intersect at two points (a, j) and (b, k) , where $j < k$. When $g(x) > f(x)$, which of the following must be true?

- A) $x > k$
 B) $x < j$
 C) $x < j$ or $x > b$
 D) $a < x < b$

32. If $\frac{2a}{b} = 4.25$ and $\frac{a}{bn} = 8.5$, what is the value of n ?

A) $\frac{1}{4}$ B) $\frac{1}{2}$ C) $\frac{1}{8}$ D) $\frac{1}{6}$

33. An object's speed is increasing at a rate of 4.80 meters per second squared. What is the rate in miles per minute squared, rounded to the nearest tenth? (Use 1 mile = 1609 meters.)

A) 10.7 B) 11.1 C) 11.8 D) 12.3

34. An acceptable noise criterion rating for the background noise in a school classroom is 30. For a noise criterion rating of 30, the equation $y = 19(0.978)^{x-50} + 37$ gives the estimated sound pressure level, y in decibels, as a function of the octave band center frequency, x , in hertz, where $x \geq 50$. Which of the following is the best interpretation of 37 in this context?

- A) 37 is 19 less than the estimated sound pressure level, in decibels, at an octave band frequency of 50.
 B) 37 is 19 less than the estimated sound pressure level, in decibels, at an octave band frequency of 0.
 C) 37 is the estimated sound pressure level, in decibels, at an octave band frequency of 50.
 D) 37 is the estimated sound pressure level, in decibels, at an octave band frequency of 0.

35. The graph of the quadratic function $y = f(x)$ in the xy -plane intersects the x -axis when $x = 74$ and $x = k$, where k is a constant. The maximum value of $y = f(x)$ occurs at the point $(17, m)$, where m is a constant. What is the value of k ?

A) -40 B) -36 C) -24 D) -32

36.

x	$f(x)$
-9	155
-3	227
3	155

Three points on the quadratic function $f(x)$ are given in the table. If $g(x) = f(x + 4)$, what is the y -intercept of the function $g(x)$?

A) 129 B) 131 C) 139 D) 141

37. Triangle ABC is an equilateral triangle with a height of 30 meters. What is the perimeter of triangle ABC in meters?

- A) $60\sqrt{3}$
 B) 90
 C) $90\sqrt{3}$
 D) $120\sqrt{3}$

38.

$$-19(6x - 4)^2 + 6(6x - 3)^2$$

The given expression can be rewritten as $\frac{a}{9}x^2 + \frac{b}{9}x + \frac{c}{9}$, where a , b , and c are constants. What is the value of $a + b + c$?

A) -198 B) -202 C) -204 D) -206

39.

A rectangular area consists of 3,179 equal squares where each square has an area of k . If the length of the rectangular area is 2.75 times the width and the width is equal to $x\sqrt{k}$, what is the value of x ?

A) 34 B) 36 C) 38 D) 42

40.

$$\begin{aligned}7x - 9y &= 39 \\ 35x - 45y &= 195\end{aligned}$$

For any real number r , which of the following points lies on the graph of both equations?

- A) $(3r, \frac{7r-13}{3})$
 B) $(\frac{7r+13}{3}, 3r)$
 C) $(-\frac{7r}{-9} + 39, \frac{7r}{9} + 195)$
 D) $(\frac{r}{5} + 39, -\frac{r}{5} + 195)$

41.

$$\begin{aligned}f(x) &= x^2 + 5x - 6 \\ g(x) &= x^2 + 2x - 35\end{aligned}$$

The quadratic function $f(x)$ has 2 solutions j and k , where $j < k$. The quadratic function $g(x)$ has 2 solutions l and m , where $l < m$. The quadratic function $h(x) = x^2 + 7x + c$ has solutions of $k + l$ and $m + j$, and can be rewritten as $(x + a)(x + b)$, what is the value of c ?

- A) 6 B) 8 C) 10 D) 12

42.

For the exponential function $f(x) = 16^{x+4}$, which of the following answer choices expresses the minimum value as either a constant or a coefficient when $x \geq 0$?

- A) $f(x) = 64(16)^x$
 B) $f(x) = 65,536(16)^x$
 C) $f(x) = 1,048,576(16)^x$
 D) $f(x) = \frac{1}{65,356}(16)^x$

43.

For the given parabola $ax^2 + bx + c$, a , b , and c are constants. The x -intercepts of the parabola are at $(14, 0)$ and $(k, 0)$. If $f(27) = f(11)$, what is the value of k ?

- A) 24 B) 28 C) 32 D) 36

44.

A quadratic function models a projectile's height, in meters, above the ground in terms of in terms of time, in seconds, after it was launched. The model estimates that the projectile was launched from an initial height of a meters and reached a maximum height of 134 meters 6 seconds after it was launched. How many seconds after launch does the model estimate it will return to a height of a meters?

- A) 6
 B) 8
 C) 9
 D) 12